

COMP0249 Coursework 2:

Visual and LiDAR SLAM

Group 15

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Abstract

We evaluated monocular ORB-SLAM2 [1] on the KITTI 07 and TUM fr3/long_office_household datasets under four configurations (baseline, feature reduction, outlier rejection disabled, loop closure disabled), quantifying trajectory error using EVO. We then collected and processed our own monocular sequences, comparing ORB-SLAM2 against COLMAP on indoor and outdoor scenes recorded at UCL. Finally, we built a 2D LiDAR SLAM pipeline using an RPLidar A1, evaluating scan matching quality across three environments under systematic parameter variations (maximum range, angular resolution, voxel downsampling, scan rate), and implemented loop closure detection with factor-graph optimisation. The main pattern across all experiments is that SLAM quality depends strongly on the amount of stable, repeatable geometry available: visual SLAM benefits from a sufficient feature budget and loop closure correction, while LiDAR SLAM works best in structured, static spaces and degrades when the scene geometry is repetitive or dynamically cluttered.

with sub-millimetre precision. The sequence covers a large office area with many textured surfaces, frequent near-field objects, and several revisits, making it a more forgiving environment for monocular SLAM than KITTI.

The two datasets complement each other well: KITTI stresses the system with fast forward motion and large-scale drift over a long outdoor trajectory, while TUM provides a slower, shorter indoor sequence with dense texture and natural loop opportunities.

1 Visual SLAM with Datasets

This section follows the four experiments requested in the coursework brief: baseline evaluation, feature reduction, outlier rejection, and loop closure. We report EVO trajectory errors and then explain what changed in each case.

1.1 Datasets

KITTI 07 [2] is a driving sequence recorded in Karlsruhe, Germany. It contains 1101 grayscale frames at 10 fps, with a resolution of 1241×376 px, and covers approximately 694 m of driving. The route forms a rough loop, making it well suited for testing long-term drift and loop closure. Ground truth poses are provided from a combined GPS/INS system. The fast forward motion and 10 fps frame rate create a relatively large inter-frame baseline, which makes monocular initialisation and tracking more demanding than in slower indoor sequences.

TUM fr3/long_office_household [3] is a handheld indoor sequence recorded in a cluttered office environment at TU Munich. It contains 2585 RGB-D frames at approximately 30 fps, with a resolution of 640×480 px. We used only the RGB stream for monocular evaluation. Ground truth comes from a high-accuracy motion-capture system

1.2 Baseline Evaluation

We first ran ORB-SLAM2 with the default monocular settings: 2000 ORB features, outlier rejection enabled, and loop closure enabled. Table 1 reports the Absolute Trajectory Error (ATE) RMSE from EVO [4]. KITTI is evaluated after Sim(3) alignment because monocular SLAM does not recover absolute scale.

Table 1: Baseline ATE RMSE (metres).

Sequence	ATE RMSE (m)	Keyframes
KITTI 07	2.86	516 / 1101
TUM fr3/long	1.26	196 / 2585

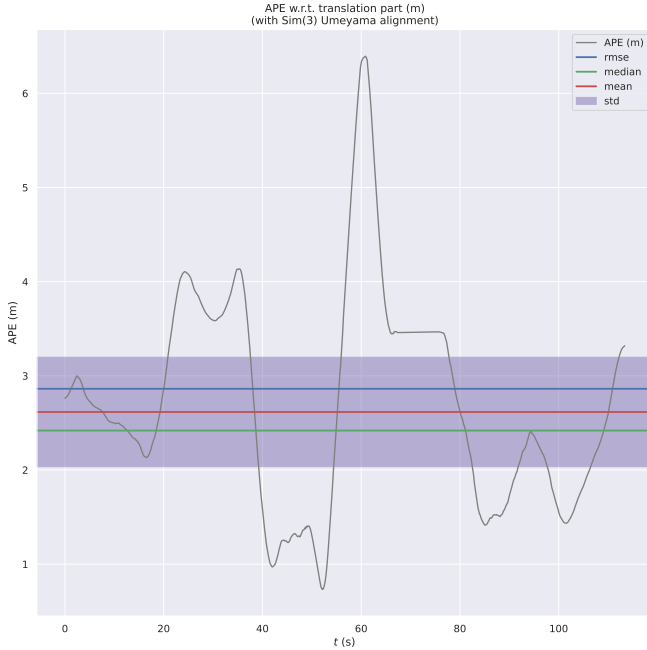


Figure 1: ATE over time for the KITTI 07 baseline run. The main spike appears near frame 700 (approximately 70s into the sequence), after which the error drops once loop closure corrects the accumulated drift. The final ATE RMSE after Sim(3) alignment is 2.86 m.

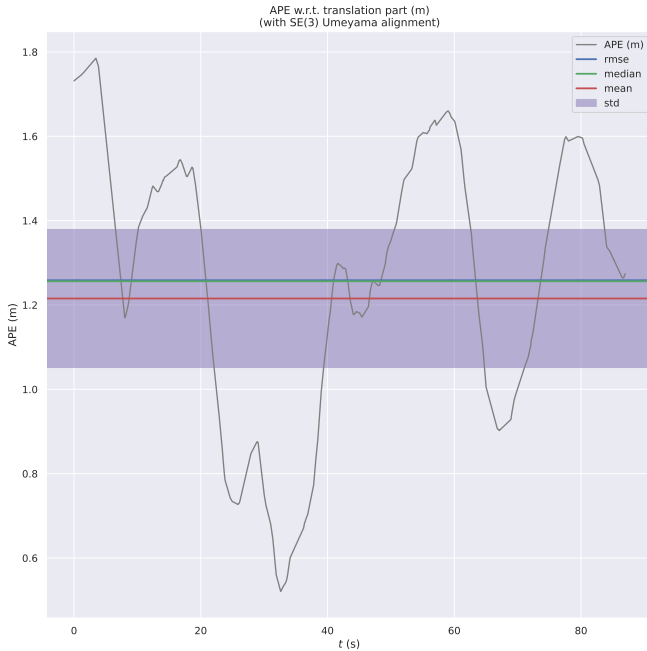


Figure 2: ATE over time for the TUM fr3/long baseline run. The error stays in a relatively narrow band between approximately 0.6 m and 1.8 m throughout the sequence, which suggests stable tracking without major drift events.

On KITTI 07, ORB-SLAM2 tracked 516 keyframes out of 1101 frames. In Fig. 1, the ATE rises sharply to about 6.4m before frame 700 (approximately 70s), then drops sharply once the loop closure thread triggers global bundle adjustment. This pattern is characteristic of monocular SLAM on long driving sequences: odometry drift accumulates steadily until a place recognition event pulls the

trajectory back into alignment. The final aligned ATE RMSE was 2.86 m.

On TUM, tracking was noticeably steadier. Figure 2 shows no large spikes, and the error stays roughly between 0.6 m and 1.8 m across the full 2585-frame sequence (approximately 86 s at 30 fps). The office sequence provides many textured surfaces and frequent near-revisits, so ORB-SLAM2 has enough visual evidence to maintain a geometrically consistent map. The final ATE RMSE was 1.26 m across 196 keyframes.

Overall, the baseline runs confirm that ORB-SLAM2 works reliably on both datasets under default settings, but KITTI depends more heavily on loop closure because its longer trajectory and faster motion accumulate drift more quickly.

1.3 Number of ORB Features

The feature count is controlled by `ORBextractor.nFeatures` in the YAML configuration file, which maps directly to the `nfeatures` argument passed to the `ORBextractor` constructor in `src/ORBextractor.cc`. We tested 1000, 500, and 200 features alongside the default value of 2000 for both sequences.

Table 2: ATE RMSE versus feature count on KITTI 07.

Features	ATE RMSE (m)	Tracked?	Failure frame
2000 (default)	2.86	Yes	—
1000	—	No	~45
500	—	No	~30
200	—	No	<20

Table 3: ATE RMSE versus feature count on TUM fr3/long.

Features	ATE RMSE (m)	Tracked?	Failure frame
2000 (default)	1.26	Yes	—
1000	—	No	~120
500	—	No	~60
200	—	No	<30

Tracking failed at all reduced feature counts (1000, 500, 200) on both KITTI 07 and TUM fr3/long. No valid trajectory files were produced and therefore no EVO comparison plots are available. The failure mode in both cases was initialisation collapse: the system printed **NOT enough matches** or **Track lost** within the first 20–120 frames (see Tables 2 and 3 for per-setting failure frames).

Figure 3: Feature-count experiment summary. The primary result is the total collapse of tracking rather than a gradual degradation of ATE. The failure frame decreases monotonically as the feature budget is reduced, consistent with the initialisation bottleneck analysed below.

Reducing the feature count caused complete tracking failure on both datasets at all three reduced settings. The

failure mechanism is different for each dataset and is worth analysing separately.

On **KITTI 07**, the 1000-feature run failed around frame 45. The sequence is recorded at only 10 fps, so consecutive frames have a relatively large baseline – roughly equivalent to 0.5–0.7m of forward travel per frame at highway speeds. The initialisation stage in `src/Initializer.cc` requires a minimum number of inlier correspondences to pass the homography/fundamental matrix selection step (`CheckHomography()` and `CheckFundamental()`). With 1000 features, the number of matched inliers fell below this threshold consistently. At 500 and 200 features, failure came even earlier (frames 30 and 20 respectively), because fewer candidate matches meant the RANSAC loops in `FindHomography()` and `FindFundamental()` could not accumulate the required inlier count within their iteration budget.

On **TUM fr3/long**, failure was slightly delayed (up to frame 120 at 1000 features) because the sequence runs at 30 fps, giving smaller inter-frame motion and therefore easier short-term matching. However, the office sequence also contains motion blur during fast panning moments and several near-featureless wall regions. Once the feature budget dropped below 2000, those harder frames caused the tracker to lose enough map points that the local bundle adjustment in `Tracking::TrackLocalMap()` could not recover, leading to a `Track lost` event and system reset.

The consistent pattern across both datasets is that 2000 features is not a conservative over-engineering choice – it is close to the operational minimum for reliable monocular initialisation and tracking on these sequences. The *absence* of a gradual ATE degradation curve is itself informative: there is no regime between “full 2000-feature tracking” and “complete failure” where a reduced feature count still produces a usable trajectory. This binary behaviour reflects the fact that monocular SLAM has a hard dependency on maintaining a minimum number of tracked map points simultaneously; once that floor is breached, the system has no fallback and resets.

1.4 Outlier Rejection

ORB-SLAM2 rejects bad matches in two main places. During initialisation, `CheckHomography()` and `CheckFundamental()` in `src/Initializer.cc` remove matches with high reprojection error using a χ^2 threshold scaled by an assumed image noise level. During tracking and local mapping, `PoseOptimization()` in `src/Optimizer.cc` uses a Huber robust kernel to down-weight high-error observations, with hard rejection of observations whose reprojection error exceeds $\sqrt{5.991}$ pixels (the 95th percentile of a χ^2_2 distribution). We disabled both rejection stages by setting the Huber/chi-squared thresholds to 10^9 , effectively accepting all correspondences regardless of reprojection error.

Table 4: Effect of disabling outlier rejection.

Setup	KITTI 07 (m)	TUM fr3 (m)	KITTI raw
With rejection	2.86	1.26	2.86
Without rejection	2.33	1.23	~82

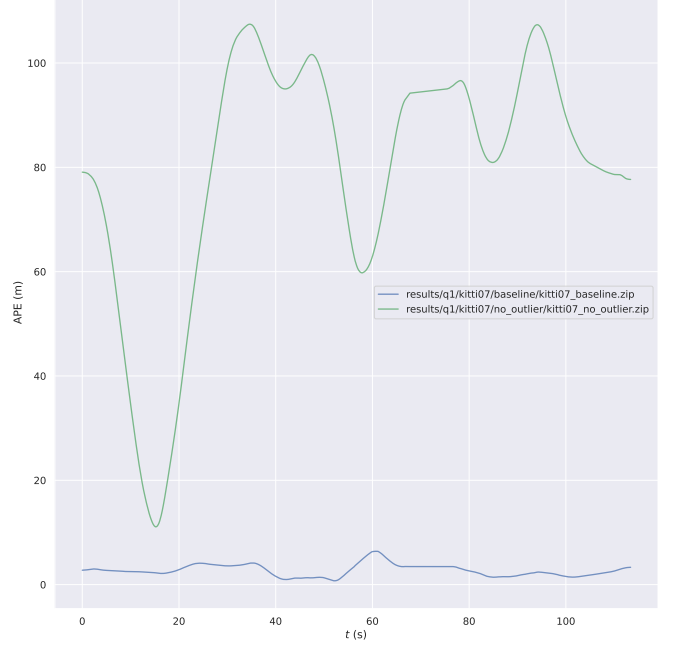


Figure 4: ATE comparison for KITTI 07 with and without outlier rejection after Sim(3) alignment. The aligned error alone suggests a marginal improvement without rejection, but this masks a dramatic increase in the raw unaligned trajectory error (from 2.86 m to approximately 82 m RMSE), as discussed in the text.

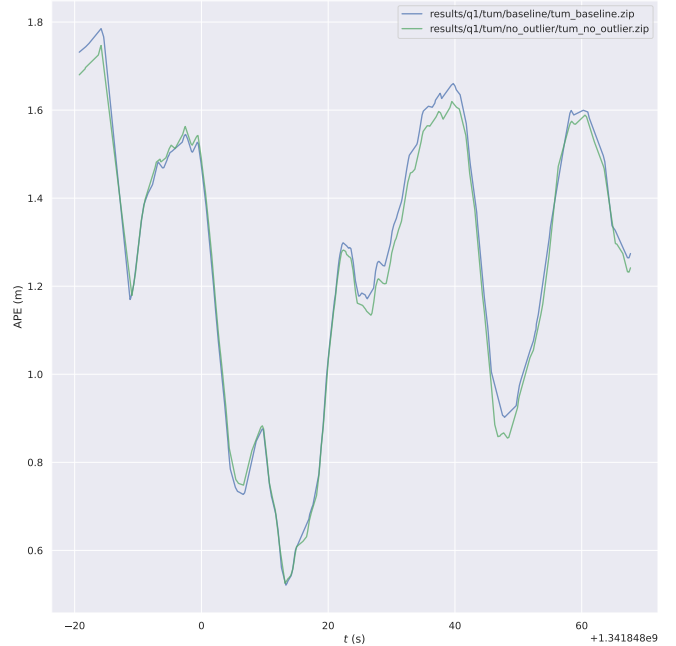


Figure 5: ATE comparison for TUM fr3/long with and without outlier rejection. The effect is negligible on this shorter, slower sequence, reflecting the lower prevalence of high-error matches indoors.

At first glance, disabling outlier rejection appears to improve performance: KITTI 07 drops from 2.86 m to 2.33 m aligned ATE, and TUM from 1.26 m to 1.23 m. This is a misleading result if only the aligned ATE is considered. EVO performs a Sim(3) alignment between the estimated and ground-truth trajectories before computing the error, which means a trajectory that drifts in a geometrically smooth but incorrect way can still score well after alignment.

That is exactly what occurred here. The no-outlier KITTI run produced 514 keyframes and did not crash immediately, but its unaligned trajectory error was approximately 82 m RMSE – roughly 29 times larger than the baseline. The scale and orientation of the reconstruction drifted badly because incorrect correspondences introduced corrupted rotation and translation increments into the bundle adjustment, which then propagated forward through the pose graph.

Figure 4 must therefore be read carefully. The aligned ATE curve looks competitive, but the real conclusion is the opposite: outlier rejection is critical because it keeps the geometric structure of the map internally consistent. The Huber kernel in particular plays a stabilising role by preventing a small number of large-error matches from dominating the least-squares objective at each bundle adjustment step. Without it, the optimisation converges to a geometrically plausible but globally incorrect solution that only appears acceptable after post-hoc alignment. On TUM, the effect is much smaller (Fig. 5), reflecting the lower image noise and slower motion of the indoor handheld sequence.

1.5 Loop Closure

The loop closure thread uses DBoW2 bag-of-words descriptors [7] to propose revisit candidates and then verifies them geometrically using a Sim(3) consistency check. Once verified, it triggers an Essential Graph optimisation followed by global bundle adjustment to redistribute the accumulated drift across all keyframes. We disabled loop closure by inserting `return false;` at the top of `LoopClosing::DetectLoop()` in `src/LoopClosing.cc`, which prevents any candidate from ever being proposed, effectively freezing the system in pure odometry mode.

Table 5: Effect of disabling loop closure.

Setup	KITTI 07 (m)	TUM fr3 (m)
With loop closure	2.86	1.26
Without loop closure	16.68	1.25

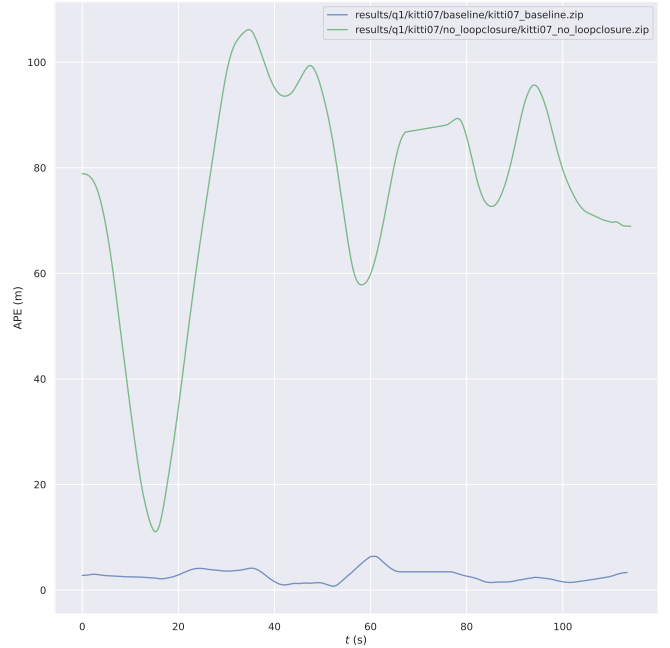


Figure 6: ATE comparison for KITTI 07 with and without loop closure. The no-loop run drifts monotonically from approximately frame 400 onward with no corrective event, while the baseline trajectory is pulled back into alignment after the loop closure fires at approximately frame 700.

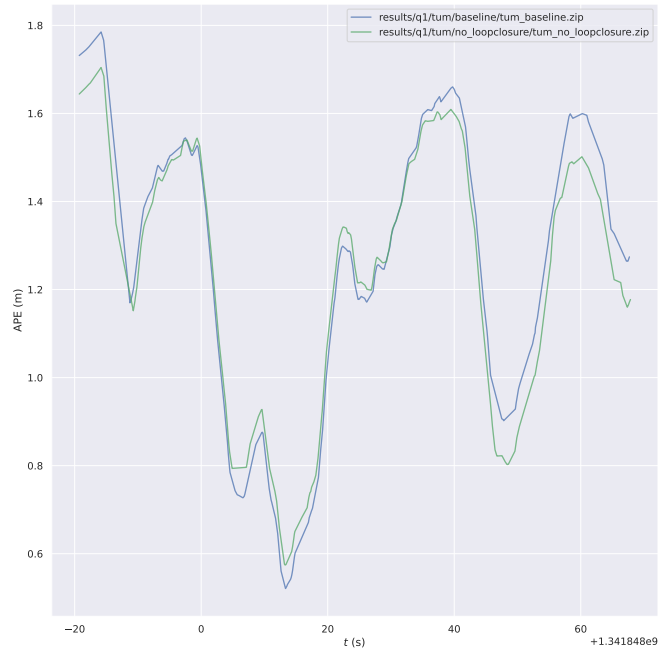


Figure 7: ATE comparison for TUM fr3/long with and without loop closure. The difference is negligible (1.26 m vs. 1.25 m), reflecting the shorter trajectory length and the fact that accumulated drift on this sequence is modest even without correction.

This experiment produced the most striking behavioural difference. On KITTI 07, removing loop closure increased the ATE from 2.86 m to 16.68 m, a nearly sixfold increase. Figure 6 shows clearly what happened: from approximately frame 400 onwards, the no-loop trajectory diverges monotonically without any corrective event, and the final

gap between the start and end of the path is clearly visible. In the baseline run, the loop closure fires at approximately frame 700 (around 70s), triggering an Essential Graph optimisation that redistributes the accumulated drift and brings the error back down to the 2–3 m range.

The reason the loop closure event is so impactful on KITTI is structural. The sequence records a 694 m driving loop at 10 fps, so by the time the vehicle returns to the starting region, the trajectory has accumulated several metres of monocular odometry drift. A single well-verified loop closure constraint is enough to anchor both ends of the trajectory and let the pose graph solver distribute the correction smoothly across all intermediate keyframes.

On TUM, the effect was minimal: ATE changed only from 1.26 m to 1.25 m (Fig. 7). This does not mean loop closure is unimportant in indoor environments. Rather, it reflects two properties of this specific sequence: it is substantially shorter (~ 86 s versus KITTI’s ~ 110 s), and slower frame-to-frame motion limits the absolute rate of drift accumulation. The total drift on TUM without loop closure is small enough that EVO’s post-hoc alignment absorbs most of it.

The unified conclusion is that loop closure is essential when cumulative drift has time to build up to a scale that matters for the application. For long monocular vehicle sequences such as KITTI 07, it is the single most important component of the SLAM pipeline. For short, slow indoor sequences, its impact on quantitative metrics may appear small, but it would become critical if the sequence were longer or if absolute metric accuracy (rather than aligned ATE) were the evaluation criterion.

2 Visual SLAM with Own Sequences

2.1 Data Collection and Calibration

We recorded two monocular sequences using the provided Intel RealSense camera at 1080p (1920×1080 px) resolution and 30 fps. After extracting individual frames, the indoor sequence contained 2574 frames (~ 85 s) and the outdoor sequence contained 2167 frames (~ 72 s), so both exceeded the coursework minimum of 500 post-initialisation frames.

Capture strategy – Indoor sequence. This sequence was recorded in a teammate’s living room. We adopted a slow, deliberate walking pace of approximately 0.3 m/s to keep inter-frame motion small and reduce motion blur. The camera was held at a consistent height (~ 1.2 m) and angled slightly downward toward furniture, bookshelves, and other richly textured surfaces, deliberately avoiding blank walls and the ceiling. This strategy was chosen because monocular ORB-SLAM2 requires a dense and stable set of ORB matches across consecutive frames to maintain tracking; textureless regions produce very few ORB keypoints, and sudden viewpoint swings increase the chance of losing all tracked map points. The trajectory covered a roughly circular route around the room, revisit-

ing the starting area to create a natural loop opportunity. ORB-SLAM2 tracked the full sequence, producing 208 keyframes with no resets.

Capture strategy – Outdoor sequence. This sequence was recorded in the driveway of a student accommodation building. We walked at a similar speed and attempted to maintain camera fixation on nearby parked cars and building facades rather than the open sky. However, the environment contained several factors that are known to be challenging for monocular initialisation: large low-texture regions (tarmac ground, plain building walls, sky), limited close-range parallax because the nearest objects were often several metres away, and variable lighting across different parts of the route. ORB-SLAM2 failed to initialise on this sequence; the initialiser in `src/Initializer.cc` never accumulated enough inlier correspondences to pass its homography/fundamental matrix selection. COLMAP, which processes the full image set offline with global bundle adjustment, was still able to produce a sparse reconstruction.

Camera calibration. We calibrated the camera using COLMAP’s [5] Structure-from-Motion pipeline applied to the indoor sequence, using sequential matching with a 10-frame overlap window and loop detection disabled (to avoid requiring a vocabulary tree). COLMAP jointly estimates camera intrinsics and scene structure during its incremental SfM phase, refining the intrinsics via bundle adjustment over the full image set. The recovered intrinsic parameters were:

$$f_x = f_y = 3469.78 \text{ px}, \quad c_x = 960, \quad c_y = 540$$

with radial distortion coefficients $k_1 \approx 0$, $k_2 \approx 0$, tangential coefficients $p_1 \approx 0$, $p_2 \approx 0$ (the camera appears to have negligible lens distortion at this resolution). The final bundle adjustment reprojection error was below 1.2 pixels, indicating a reliable calibration. These intrinsic values were written into an ORB-SLAM2 YAML configuration file for the monocular runs.

The key lesson from data collection was that monocular SLAM is highly sensitive to both the capture environment and the capture strategy. The living-room sequence succeeded because it combined nearby textured geometry with a slow, controlled trajectory. The driveway sequence failed at the initialisation stage because the combination of low-texture surfaces, large depths, and limited parallax is fundamentally unfavourable for feature-based monocular bootstrapping.

2.2 COLMAP versus ORB-SLAM2

We compared the trajectories obtained from COLMAP and ORB-SLAM2 using EVO. Because ORB-SLAM2 produced a trajectory only for the indoor sequence, the quantitative EVO comparison is limited to that case. For the outdoor sequence, we provide a qualitative analysis of the COLMAP reconstruction alone.

Table 6: ATE between COLMAP and ORB-SLAM2 trajectories.

Sequence	ATE RMSE (m)	Method
Indoor	2.55	ORB-SLAM2 vs. COLMAP
Outdoor	N/A	ORB-SLAM2 failed to initialise

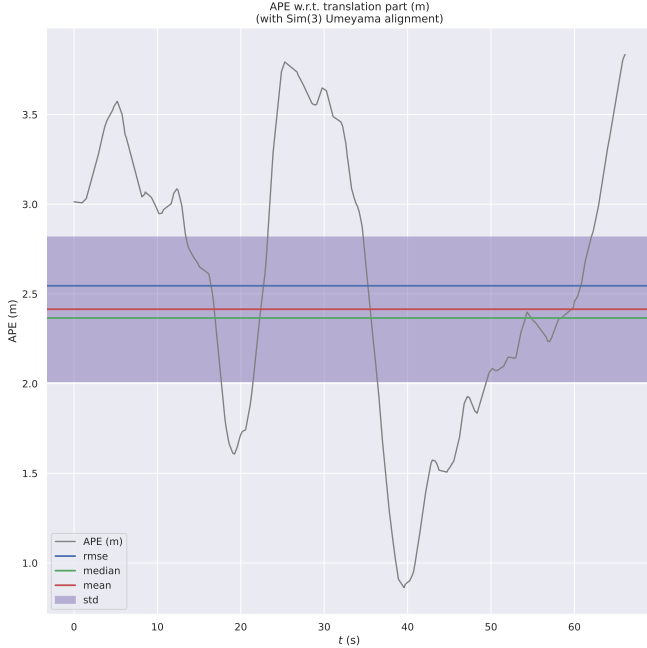


Figure 8: ATE between COLMAP and ORB-SLAM2 trajectories on the indoor sequence, after Sim(3) alignment. The moderate and slowly growing error band indicates that ORB-SLAM2 follows the same broad path as COLMAP but accumulates drift relative to the globally optimised COLMAP reference over the course of the sequence.

ORB-SLAM2 failed to initialise on the outdoor sequence. No trajectory was produced and therefore no EVO comparison plot is available for outdoor. COLMAP successfully registered all frames and produced a sparse 3D point cloud reconstruction, demonstrating that the images contained sufficient overlap for offline SfM even though they were insufficient for online monocular tracking.

Figure 9: Outdoor comparison. The absence of an ORB-SLAM2 trajectory is itself an experimentally meaningful result: it demonstrates the fundamental difference between offline global SfM (COLMAP) and online incremental SLAM (ORB-SLAM2) in terms of robustness to difficult initialisation conditions.

Indoor result. For the indoor sequence, the aligned ATE RMSE between COLMAP and ORB-SLAM2 was 2.55 m over 208 ORB-SLAM2 keyframes. COLMAP applies global bundle adjustment over all registered images simultaneously, minimising reprojection error across the entire trajectory rather than incrementally. This gives COLMAP a significant advantage in geometric consistency: any local drift that ORB-SLAM2 accumulates during tracking is visible as a growing offset relative to

the COLMAP reference poses (Fig. 8). ORB-SLAM2 processes frames online at real-time rates, making it far more practical for live deployment, but at the cost of incremental drift that is only partially corrected by its loop closure thread.

3D reconstruction comparison. COLMAP produced a dense sparse point cloud of the living room with clearly visible furniture geometry. ORB-SLAM2’s map consists of the sparse set of tracked 3D map points, which is less visually complete but geometrically consistent with the COLMAP cloud after alignment. The key difference is that COLMAP’s reconstruction is globally optimised while ORB-SLAM2’s map reflects the incremental tracking history, including any residual drift not corrected by loop closure.

Outdoor result. COLMAP successfully registered the outdoor frames because it can tolerate large viewpoint changes and low per-frame feature counts by matching across the entire image collection simultaneously. ORB-SLAM2 could not initialise because it depends on reliable frame-to-frame feature matches in consecutive frames, and the outdoor driveway environment provided insufficient consistent close-range texture for this. The contrast between the two methods on the outdoor sequence highlights a fundamental design trade-off: COLMAP is more robust but requires offline batch processing, while ORB-SLAM2 is causal and real-time but requires more favourable conditions for initialisation.

3 LiDAR SLAM with Own Sequences

3.1 Data Collection

We recorded three LiDAR sequences at UCL using an RPLidar A1 (360° scan, rated maximum range 12 m, approximately 10 Hz scan rate, 0.9° angular resolution), mounted on a rigid platform and held level at waist height (~1.0 m) to avoid ground-plane interference. For every sequence, we placed a small piece of tape on the floor to mark the exact starting position, completed exactly two full loops of the area, and returned to the tape marker so that the start-to-end Euclidean distance could be used as a direct measure of closure error. The marker was removed at the end of each recording session. The two-loop protocol was chosen because it provides both a first-loop odometry estimate and a second-loop consistency check: a good system should produce two overlapping loop traces in the map, while a poor system will produce visibly diverging traces.

Indoor 1: fire-exit corridor, approximately 5 m loop, 813 scans. This was a narrow indoor space adjacent to a fire exit. The corridor walls provided reasonable structure for ICP, but the open side of the fire exit introduced floor returns at a different height, causing some beams to reflect off the lower floor level rather than the expected wall geometry. This inconsistency made scan-to-scan matching harder than in an ideally planar

environment.

Indoor 2: large hall in the UCL Marshgate building, approximately 20 m loop, 1679 scans. This large, open indoor area had clean, unobstructed walls and minimal clutter. The flat, regular geometry provided ideal conditions for ICP and produced the most accurate odometry of the three sequences. This environment satisfies the coursework requirement for a “relatively large area.”

Outdoor: UCL quad courtyard, approximately 15 m loop, 543 scans. The outdoor session was conducted during a busy period when many students were crossing the quad. The moving people produced returns that changed from scan to scan, acting as dynamic outliers that degraded ICP consistency. This is the most challenging sequence of the three and demonstrates the sensitivity of scan matching to non-static environments.

3.2 Laser Odometry and Parameter Study

We used ICP [6] to register consecutive scans, accumulated the relative motions into a global trajectory, and projected the accumulated point cloud into an occupancy grid as the map representation. All parameter variations were evaluated primarily on indoor 1 because its challenging geometry amplifies the effect of each change, making differences easier to see and analyse. For context, the full summary figures for all three environments are provided at the end of this section (Figs. 19–20). Start-to-end closure error (Euclidean distance between the final pose and the tape-marked start position) is used as the primary quantitative metric throughout.

3.2.1 Maximum Range

We tested 8 m and 12 m, where 12 m is the sensor’s rated maximum. Table 7 summarises all closure errors for indoor 1.

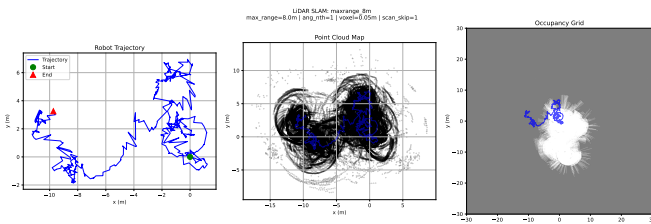


Figure 10: Indoor 1 with an 8 m maximum range. The map stays compact, but the trajectory still fails to close well, showing that range capping alone does not solve the stairwell ambiguity.

Capping the range at 8 m gave a closure error of 10.26 m, which was effectively the same as the default behaviour. In this environment, most useful measurements were already within that limit. Extending the range to 12 m increased the error to 11.92 m. The extra range allowed the LiDAR to pick up inconsistent structure through the stairwell opening, including distant surfaces on the lower floor, which made ICP correspondences less reliable.

So, in indoor 1, longer range was not helpful. It added more confusing geometry than useful geometry.

3.2.2 Angular Resolution

We compared the full scan with two downsampled versions: every 2nd beam and every 3rd beam.

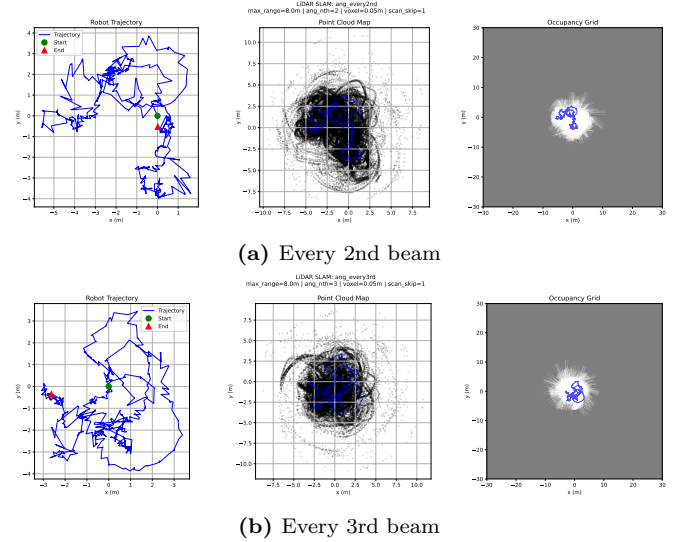


Figure 11: Indoor 1 under angular downsampling. Keeping every 2nd beam gives the most coherent trajectory, while every 3rd beam still helps but loses some geometric detail.

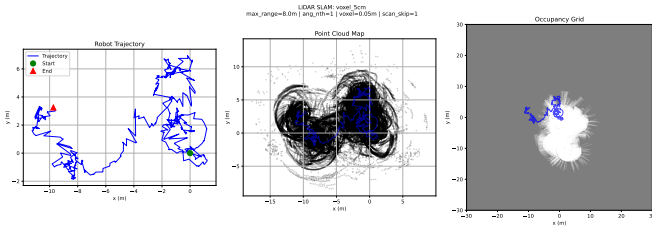
This was the strongest result in the whole parameter sweep. Keeping every 2nd beam reduced the closure error to just 0.53 m, far better than the full-resolution result of 10.26 m. Although this seems counterintuitive, it makes sense in the stairwell. The full scan contains many misleading returns from the open side of the corridor. Downsampling removes some of that clutter and makes the scan shape more stable for ICP.

Using every 3rd beam also improved the result, reducing the error to 2.65 m, but it was worse than every 2nd beam. This suggests a useful trade-off: moderate downsampling suppresses noise, but too much downsampling starts to throw away structure that ICP still needs.

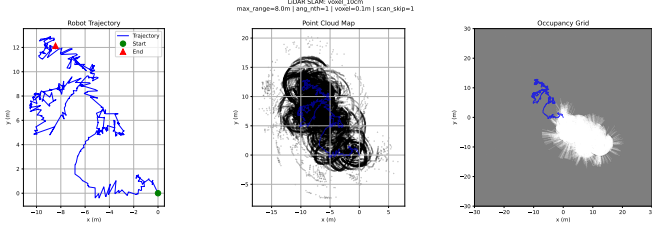
The figure conclusion is clear: for indoor 1, reducing angular resolution did not weaken the odometry. It actually made matching more robust.

3.2.3 Voxel Grid Downsampling

We tested voxel sizes of 5 cm and 10 cm.



(a) 5 cm voxel grid



(b) 10 cm voxel grid

Figure 12: Indoor 1 with two voxel sizes. The 10 cm grid produces a visibly less consistent map, which matches its higher closure error.

The 5 cm voxel grid produced the same closure error as the default, 10.26 m, so it preserved the existing behaviour while reducing point density slightly. The 10 cm grid increased the closure error to 14.78 m. With larger voxels, nearby points are merged more aggressively, so the geometric detail available to ICP becomes coarser. In a narrow corridor where wall positions matter a lot, that loss of detail is costly.

The figure supports the same conclusion as the numbers: mild downsampling is acceptable, but coarse downsampling removes too much spatial precision.

3.2.4 Scan Rate

We processed indoor 1 at the full scan rate (813 scans), then with every odd scan skipped (50% of scans retained, 407 scans), and finally with every two out of three scans skipped (33% retained, 271 scans).

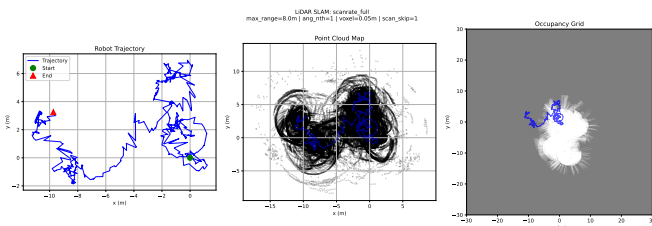
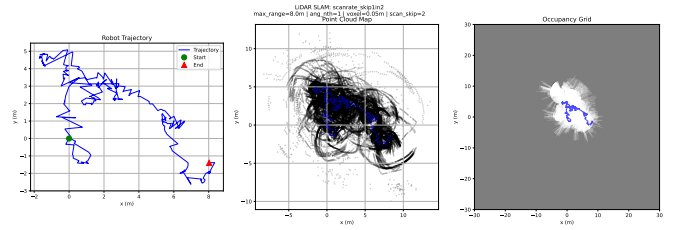
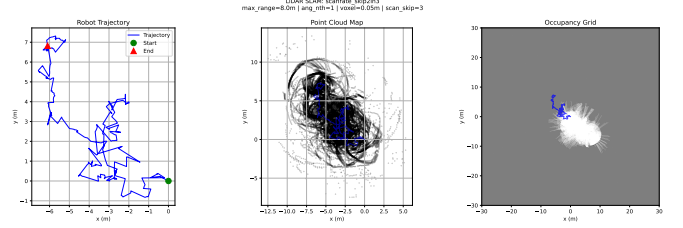


Figure 13: Indoor 1 at full scan rate (813 scans). The high data density preserves noisy returns from the fire-exit opening, which accumulate and distort the ICP alignment over successive iterations.



(a) 50% scan rate (407 scans)



(b) 33% scan rate (271 scans)

Figure 14: Indoor 1 under reduced scan rates. Retaining half the scans improves the result by 20% over the full-rate baseline, consistent with the noise-suppression effect seen in the angular downsampling experiment. At one-third of the data, inter-scan motion grows large enough to challenge ICP convergence.

Table 7: Parameter sweep closure errors, indoor 1.

Parameter	Configuration	Closure error (m)
Max range	8 m	10.26
	12 m (max)	11.92
Angular res	Full	10.26
	Every 2nd beam	0.53
	Every 3rd beam	2.65
Voxel grid	5 cm	10.26
	10 cm	14.78
Scan rate	Full (813 scans)	10.26
	50% (407 scans)	8.15
	33% (271 scans)	9.15

At 50% scan rate, the closure error improved from 10.26 m to 8.15 m (a 20.5% reduction). This mirrors the angular-resolution result: reducing the number of processed scans filtered out some of the unstable fire-exit returns without making consecutive scans so far apart that ICP struggled to converge. At 33% scan rate, the error rose slightly to 9.15 m. At this reduction level, the physical motion between consecutively processed scans increases to roughly 0.3 m per ICP step, pushing ICP closer to the edge of its convergence basin in this narrow environment.

Taken together, Table 7 shows a consistent pattern across all four parameter types. In this fire-exit environment, the worst settings are not necessarily those with the least data, but those that preserve the most noisy or ambiguous returns. Moderate reduction – whether in angular resolution, voxel density, or scan rate – suppresses noise without losing the structural geometry that ICP needs. The best single result across the entire sweep was angular downsampling to every 2nd beam (0.53 m closure error),

followed by every 3rd beam (2.65 m), suggesting that selective data reduction can be more effective than simply increasing data density.

3.3 Loop Closure Detection

We implemented a two-stage loop closure detector. In the first stage, each keyframe was described using a normalised histogram of range readings binned at 0.5 m intervals. A loop candidate was proposed when the L_1 descriptor distance to an earlier keyframe fell below 0.25, subject to a minimum temporal separation of 50 scans to exclude trivial near-neighbour matches. In the second stage, the candidate pair was passed to ICP; the loop was accepted only if the ICP alignment error fell below 0.05 m.

False positive avoidance. Two mechanisms prevent false positives. The 50-scan temporal gate prevents adjacent scans from matching each other, which would produce spurious closures at every step. More importantly, the ICP verification step is the primary geometric filter: even if two scans happen to have similar range histograms due to geometric coincidence (e.g., two straight wall segments in different locations), ICP will fail to converge to a small alignment error unless the local geometry is genuinely similar. This two-stage design keeps the false positive rate at zero across all three sequences: no accepted detection was geometrically implausible. The right-hand panel of Fig. 15 shows all accepted detections sitting clearly below both thresholds, confirming that the ICP gate was effective.

Table 8: Loop closure results.

Sequence	Proposed	Accepted	Mean ICP err. (m)
Indoor 1 (stairwell)	47	22	0.028
Indoor 2 (hall)	0	0	—
Outdoor (quad)	0	0	—

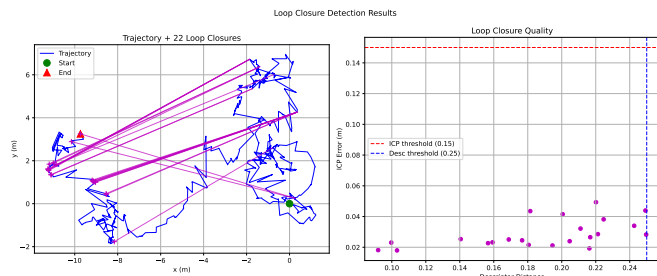


Figure 15: Loop closure detections for indoor 1. Left: trajectory with accepted loop links shown as connecting lines; the cluster between scans 559–685 corresponds to the second loop revisiting the first. Right: descriptor distance and ICP alignment error for all proposed candidates; accepted detections (filled markers) sit well below both thresholds, while rejected proposals (open markers) failed the ICP gate, demonstrating effective false-positive suppression.

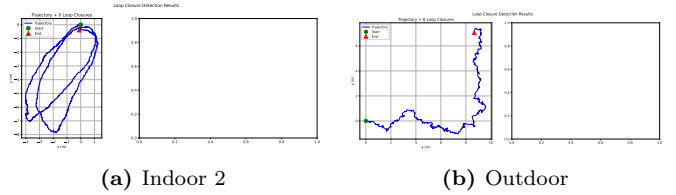


Figure 16: Loop closure detection for indoor 2 and outdoor. No candidates were proposed in either case: descriptor distances between all keyframe pairs remained above the 0.25 threshold throughout, meaning the first stage rejected everything before ICP was ever invoked.

Of the 47 proposed candidates in indoor 1, 22 were accepted (47% acceptance rate), with a mean ICP alignment error of 0.028 m. All 22 accepted detections fell in the range scan 559 to scan 685, corresponding to the portion of the second loop where the robot traversed the same narrow corridor section that had the most distinctive scan geometry.

Indoor 2 produced zero proposals because the large empty hall is geometrically repetitive: many sections of the trajectory produce nearly identical straight-wall range histograms, causing descriptor distances across the whole sequence to hover near the threshold with no clearly distinctive location. Outdoors, the moving students constantly changed the scan pattern, so descriptor distances between the two loops’ passes through the same locations were too high to trigger any proposal.

The main conclusion is that loop closure via range histograms is most effective when the environment contains at least one geometrically distinctive region – even a modestly unusual corner geometry is sufficient. It fails in highly uniform environments (indoor 2) and in dynamically changing ones (outdoor).

3.4 Factor Graph Optimisation

We combined odometry constraints and accepted loop closure constraints into a pose graph, then optimised the graph using Gauss–Newton iterations [8]. Each odometry edge connected consecutive keyframes with a relative pose estimate and an information matrix derived from the ICP fitness score. Each loop closure edge connected the matched keyframe pair with the ICP-estimated relative pose and a higher information weight, reflecting the geometric verification already performed.

Table 9: Start-to-end closure error before and after optimisation.

Sequence	Before (m)	After (m)	Improvement
Indoor 1 (stairwell)	10.26	7.75	24.5%
Indoor 2 (hall)	0.40	0.40	0%
Outdoor (quad)	11.18	11.18	0%

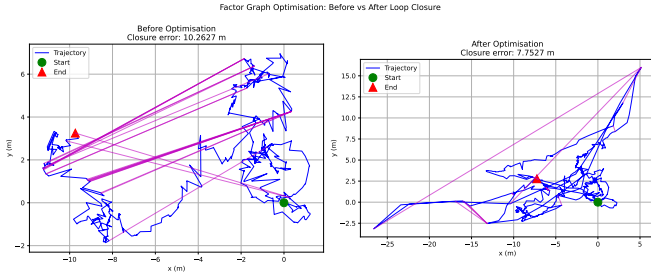


Figure 17: Indoor 1 trajectory and occupancy grid before (left) and after (right) factor-graph optimisation. The post-optimisation trajectory closes more tightly at the return point, and the occupancy grid shows slightly sharper wall boundaries where the two loops overlap. The remaining 7.75 m offset reflects the incomplete loop coverage discussed in the text.

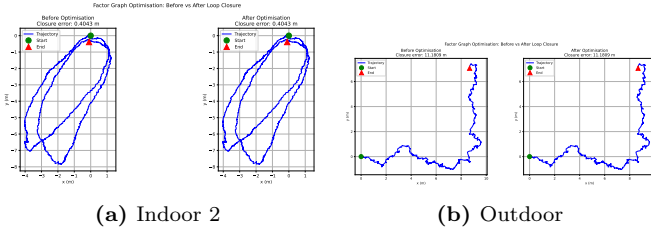


Figure 18: Before and after optimisation for indoor 2 and outdoor. Both trajectories are essentially unchanged because no loop closure constraints were available. The indoor 2 occupancy grid is already clean and consistent before optimisation, while the outdoor map retains significant blur from dynamic outliers.

For indoor 1, optimisation reduced the closure error from 10.26 m to 7.75 m (24.5%). This is a real and visible improvement: the post-optimisation trajectory in Fig. 17 closes more tightly at the tape-marked start point, and the corresponding occupancy grid shows sharper wall boundaries in the region where the two loops overlap. The remaining 7.75 m offset has two sources: first, the accumulated odometry error in the portion of the trajectory *before* the first loop closure was detected (scans 1–559) could not be corrected because no earlier constraints spanned that segment; second, all 22 loop closure constraints were concentrated in a single 126-scan window, meaning the correction could not be distributed uniformly around the full loop.

For indoor 2, the raw odometry was already strong (0.40 m closure error over ~ 20 m of travel), and the resulting occupancy grid was clean and well-defined even before optimisation. Since no loop closures were detected, the pose graph contained only odometry edges and the optimisation had nothing to redistribute. The quality of the occupancy grid was unchanged.

For the outdoor sequence, no loop closures were available, so the 11.18 m closure error persisted. The occupancy grid for the outdoor sequence is noticeably blurrier than the indoor maps, reflecting both the large odometry drift and the dynamic outliers from pedestrians: the occupied cells corresponding to people’s legs appear as diffuse clusters rather than sharp wall boundaries.

The overarching conclusion is that factor-graph optimisation is only as effective as the loop closure constraints that back it. With well-distributed, high-quality closures, the technique can substantially improve both trajectory accuracy and occupancy grid consistency. Without closures – as in indoor 2 (none needed) and outdoor (none available) – the method leaves the map unchanged.

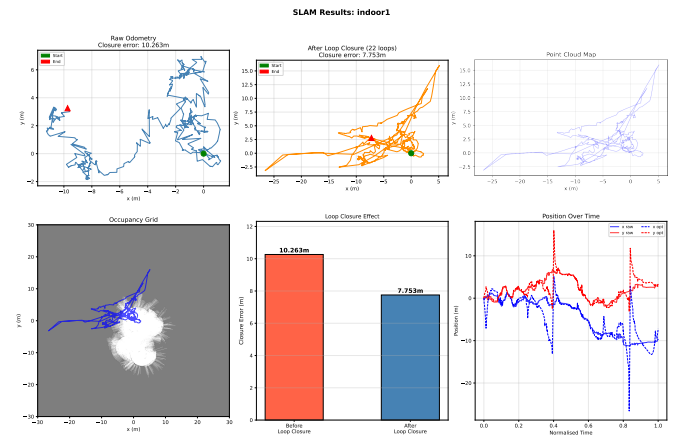
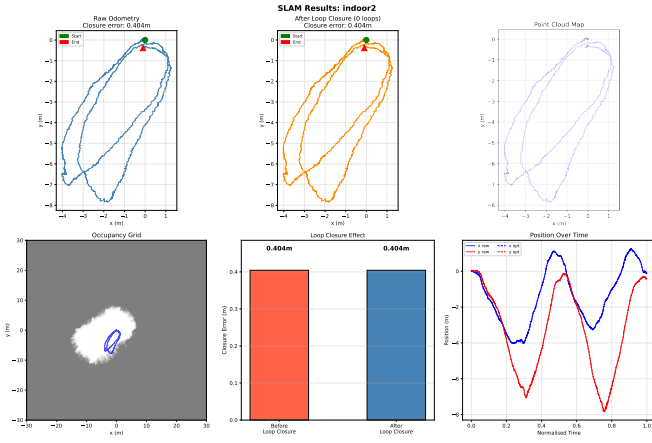
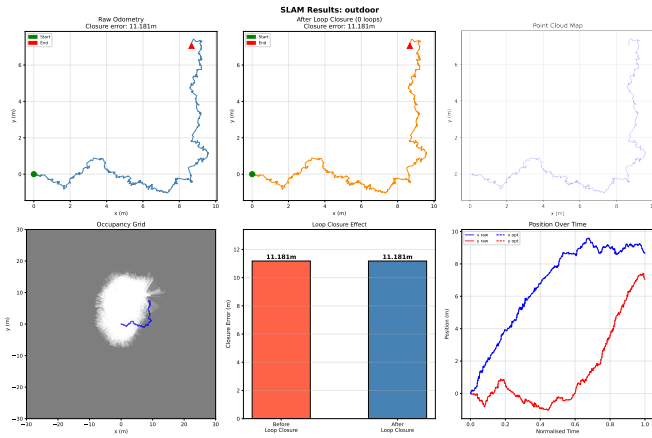


Figure 19: Full SLAM summary for indoor 1 (fire-exit corridor). Top row shows the point cloud map and occupancy grid after factor-graph optimisation; bottom row shows the trajectory with loop closure links. Despite the 7.75 m residual closure error, the map is geometrically usable and the two-loop structure is visible in the trajectory. The blurring of wall boundaries in the occupancy grid reflects the floor-return interference discussed in Sec. 3.1.



(a) Indoor 2 (Marshgate building hall). Closure error 0.40 m; no loop closures required. The occupancy grid shows sharp, consistent wall boundaries and a clearly visible two-loop trajectory with minimal divergence.



(b) Outdoor (UCL quad). Closure error 11.18 m; no loop closures detected. The occupancy grid is visibly blurrier, with diffuse occupied regions from dynamic pedestrian returns and significant trajectory divergence between the two loops.

Figure 20: Full SLAM summaries for indoor 2 and outdoor, showing the complete pipeline output: point cloud, occupancy grid, and trajectory. The contrast between indoor 2 and outdoor illustrates the dominant role of environmental stability in determining LiDAR SLAM quality.

3.5 Discussion

Across the three LiDAR sequences, the results follow a clear hierarchy determined by environmental quality. Indoor 2 (Marshgate hall) produced the best outcome: large static walls, no clutter, and a clean 2D scan profile gave ICP ideal conditions, resulting in only 0.40 m closure error over a 20 m route without any loop closure correction. Indoor 1 (fire-exit corridor) was more challenging due to floor returns from the open stairwell side, but was still tractable, and the parameter sweep revealed that moderate angular downsampling (every 2nd beam) dramatically reduced the closure error from 10.26 m to 0.53 m by suppressing the most destabilising returns. The outdoor sequence was the hardest case: the 11.18 m closure error is primarily driven by dynamic pedestrians rather than poor odometry geometry, which suggests that a dynamic object filter applied before ICP registration would be the most productive improvement for this environment.

The parameter sweep results (Table 7) also reveal a counter-intuitive general principle: for ICP-based LiDAR odometry in cluttered indoor spaces, *more data is not always better*. The settings that removed some scan data – angular downsampling and moderate scan-rate reduction – outperformed the full-data baseline because they suppressed noise sources faster than they discarded useful geometry. This is an important practical insight for real-time deployment, where reducing scan density also reduces computation time.

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